

PD Peritonitis

Diagnosis, empiric therapy, catheter decisions

Cloudy effluent = peritonitis until proven otherwise. Culture FIRST, then IP antibiotics.

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MS3/MS4 rotation teaching

Why we get consulted

PD patient with cloudy effluent, pain, or fever.

CONSULT TRIGGERS

- Cloudy effluent
- Abdominal pain in PD patient
- Fever / leukocytosis
- Exit-site or tunnel infection
- Refractory at 48–72 h
- Relapsing / recurrent peritonitis
- Fungal — urgent catheter decision

TEACHING PEARL

*Send effluent in blood culture bottles
BEFORE antibiotics (much higher yield).
Then start empiric IP therapy. Treat on
clinical suspicion + cloudy.*

Diagnosis: the 2-of-3 rule

Make the diagnosis when any 2 of 3 criteria are met.

1

CLINICAL FEATURES

Abdominal pain and/or cloudy PD effluent. Pain is usually diffuse; rebound + rigidity = surgical abdomen until proven otherwise.

2

EFFLUENT WBC

> 100 cells/ μ L (with at least 2-h dwell time) AND > 50% polymorphonuclear leukocytes (PMNs). In short dwell (automated peritoneal dialysis (APD)), weight the PMN % heavily.

3

POSITIVE CULTURE

Organism isolated from PD effluent. Yield is higher if sent in blood-culture bottles (ISPD recommended). Culture-negative peritonitis ~15% (often pre-antibiotic draw).

What to send — in order

Effluent BEFORE antibiotics. Then start IP therapy empirically.

- 1 Drain-in-drain-out: send first drained bag effluent (minimum 50 mL), centrifuge 3000×g × 15 min.
- 2 Inoculate sediment into blood-culture bottles (aerobic + anaerobic). Also send Gram stain, cell count with diff.
- 3 Exam: exit site (erythema, pus, crust), tunnel palpation, abdomen (diffuse vs localized, rebound, rigidity).
- 4 Blood cultures if febrile; CBC, BMP, lactate, bowel imaging (CT) if surgical abdomen.
- 5 Start empiric IP antibiotics AFTER cultures sent (see next slide). Do not delay for lab results.

Empiric IP antibiotic regimens

Cover gram-positive + gram-negative; intraperitoneal, not IV.

Regimen	Dose (intraperitoneal)	Notes
Vancomycin + 3rd-gen cephalosporin	Vanc: 15–30 mg/kg IP load q5–7 d (redose per level) Ceftazidime: 1 g IP daily	Most common regimen; watch vanc level; covers MRSA + enteric gram-negative rod (GNR)
Cefazolin + ceftazidime	Cefazolin: 15–20 mg/kg IP daily Ceftazidime: 1 g IP daily	Alternative if low MRSA prevalence; cheaper
Vancomycin + aminoglycoside	Vanc as above Gentamicin 0.6 mg/kg IP daily	Avoid prolonged courses (ototoxicity, nephrotoxicity in residual function)
Cefepime monotherapy	1 g IP daily	ISPD 2022 acceptable alternative; broad coverage

Why IP not IV: much higher peritoneal concentration; avoids systemic exposure. Continuous dosing in every dwell OR intermittent once-daily dosing; fill volume at normal dwell time. Add heparin 500 U/L to prevent fibrin clots.

Tailoring therapy

Narrow at 48–72 h based on culture + sensitivities. Total course 14–21 days.

Organism	Preferred therapy	Duration
CoNS (coag-neg staph)	Cefazolin or vanc IP (per susceptibility)	14 days
<i>S. aureus</i> (MSSA / MRSA)	Cefazolin (MSSA) / vanc (MRSA); + rifampin if exit-site source	21 days
<i>Streptococcus</i> / <i>Enterococcus</i>	Ampicillin ± gent (enterococcus); vanc if resistant	14 days (strep) / 21 days (enteroc.)
Single GNR (<i>E. coli</i> , <i>Klebsiella</i>)	Ceftazidime or cefepime per sensitivities	21 days
<i>Pseudomonas</i>	Dual coverage (ceftaz + gent or cipro); often removes catheter	21 days
Polymicrobial / enteric	Think surgical abdomen — CT, surgery consult; broad-spectrum IV + IP	Variable; often catheter removal
Fungal	Remove catheter IMMEDIATELY; IV fluconazole or echinocandin	≥ 2 wk after catheter removal
<i>Mycobacteria</i>	Rare; often remove; 4-drug ID-directed regimen	ID-guided

When to remove the catheter

Removal is common; learn the triggers early.

REMOVE CATHETER

Fungal peritonitis — immediately

Refractory peritonitis (no improvement by day 5 of appropriate IP abx)

Relapsing peritonitis (same organism within 4 wk of course end)

Mycobacterial peritonitis — usually

Concurrent tunnel infection with same organism

Polymicrobial with enteric flora (think surgical perforation)

Refractory exit-site / tunnel infection

KEEP CATHETER

Clinical improvement + effluent clearing by 48–72 h

Single organism susceptible to narrow therapy

No tunnel / exit-site involvement

Effluent WBC trending down even if still > 100

ISPD 2022: if effluent WBC clearly improving, continuing abx beyond day 5 is reasonable before removal

After removal: minimum 2-week antibiotic course + a few weeks on HD before new PD catheter

Prevention

Training, exit-site care, prophylaxis for specific situations.

EXIT-SITE CARE

Daily cleansing with chlorhexidine or antibacterial soap. Topical mupirocin or gentamicin cream at exit site daily (proven to reduce peritonitis). Avoid immersion in swimming pools / hot tubs during catheter placement.

TRAINING + RETRAINING

Initial training > 15 hours; home visits after 2 weeks. Retrain after every peritonitis episode — poor technique is the #1 reversible risk factor. Assess hand hygiene, bag exchange technique, exit-site care.

PROCEDURE PROPHYLAXIS

Colonoscopy or invasive GYN procedures: antibiotic prophylaxis + empty abdomen per center protocol. Review any planned invasive procedure with the PD team.

CONSTIPATION + HYPOKALEMIA

Constipation → gram-negative peritonitis; aggressive stool management. Hypokalemia → gut bacterial translocation, higher peritonitis risk — replete.

Common pitfalls

The errors that cause treatment failure or catheter loss.

AVOID

- Waiting on culture to start abx
- IV antibiotics instead of IP
- Effluent in plain tubes
- Skipping exit-site exam
- Keeping catheter for fungal
- Refusing to remove refractory catheter
- No mupirocin prophylaxis

DO INSTEAD

- Send effluent + start IP abx together
- IP dosing (continuous or daily)
- Always blood-culture bottles
- Exam exit site + tunnel every visit
- Remove for fungal, refractory, relapsing
- Day 5 no improvement → remove
- Daily exit-site mupirocin

Clinical case

Think it through — answer is on the next slide.

VIGNETTE

A 52-year-old man on CAPD $\times$ 3 yr for ESRD from DKD, presents with 1 day of cloudy effluent and abdominal pain. Last episode 8 months ago (staph epi). T 37.9, BP 128/78. Abdomen: diffuse tenderness, no rigidity. Exit site: clean, no erythema. Effluent sent: WBC 1400/ μ L, 82% PMN. Gram stain: gram-positive cocci in clusters. Cultures pending.

QUESTION: What's the empiric regimen, and what would trigger catheter removal?

Clinical case — answer

Key teaching points from the case.

ANSWER

IP vancomycin (15-30 mg/kg load) + IP ceftazidime (1 g daily); empiric coverage until cultures final. Remove catheter for fungal, refractory (no improvement by day 5), or relapsing peritonitis.

TEACHING

Classic PD peritonitis — cloudy effluent + pain + effluent WBC > 100 with > 50% PMN meets 2-of-3 ISPD criteria. Always send effluent in blood-culture bottles BEFORE antibiotics. Gram-positive on Gram stain suggests staph — continue vanc until sensitivities back. Dual coverage (add ceftaz) for gram-negative until final cultures. Duration 14-21 days (21 for staph aureus). Keep the catheter unless: fungal (remove immediately), refractory at day 5, relapsing (same org within 4 wk), mycobacterial, concurrent tunnel infection. After the episode: review exit-site technique, re-training (poor technique is #1 reversible risk), continue mupirocin at exit site daily.

REFERENCES

PD Peritonitis

Guidelines

ISPD 2022 Peritonitis Guidelines

ISPD 2016 Peritoneal Dialysis-Related Infections

ISPD 2017 Catheter-related infections

KDOQI / ISPD PD Adequacy Guidelines

Landmark Trials

Perez Fontan et al. — mupirocin exit-site prophylaxis

Bernardini et al. — gentamicin cream vs mupirocin

PD Outcomes Registry data (USRDS, BRAZPD, etc.)

Ballinger et al. — rapid cycler peritonitis outcomes

UpToDate (2026 access):

Clinical manifestations and diagnosis of peritonitis in peritoneal dialysis • Treatment of peritonitis in peritoneal dialysis • Exit-site and tunnel infections in peritoneal dialysis • Peritoneal dialysis catheter-related infections: Prevention